Introduction To Programming

Python fundamentals

Universidad...

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Today's Goals

- Fundamentals
- Type of Variables
- Your first program

Fundamentals

Computers are obedient, not smart. They do exactly what you tell them to do, word for word, step by stepeven if your instructions lead to a error.
An **algorithm** is a precise, step-by-step set of instructions designed to solve a specific problem or complete a task.

Fundamentals

Computers Do What You Tell Them.

Think a computer of a kitchen assistant.If you tell it to "cook a complex dish". It will freeze because it doesn't know what that means. But if you give it a step-by-step recipe like: "chop 5 tomatoes, blend for 10 seconds, add 2 grams of salt", it will do it very quickly; because it moves at millions of operations per second.

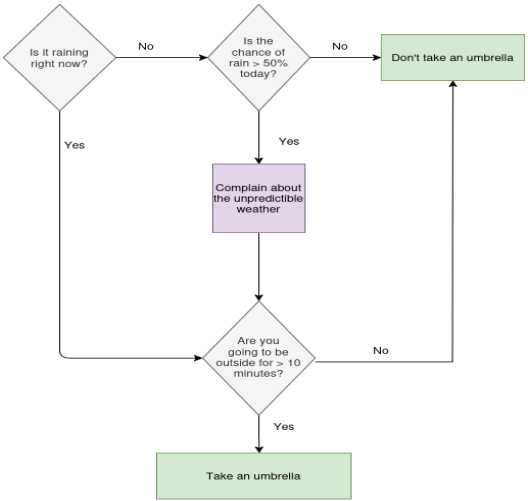
You Should Be Familiar With...

The Concept Of:

- ◇ Variable
- ◇ True/false statements
- ◇ Connecting statements (or,and, not)
- ◇ Making decisions (if ... then)
- ◇ Following flowcharts

If you shouldn't be familiar with the Flowcharts

You should be familiar with following a set of instructions and making branching desicions.



The structured program theorem

States that any computable algorithm can be expressed only with three logic control structures.

- Sequence or simple action: Executing one instruction after another in a specific, linear order.
- Conditional: valuating a boolean condition to "choose the correct path".
- Loop or iteration: Repeating a block of instructions as long as a certain condition remains true.